



ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY
Guwahati
Course Structure and Syllabus

(From Academic Session 2024-25 onwards)

B. TECH
INSTRUMENTATION ENGINEERING
3rd SEMESTER



ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY

Course Structure (From Academic Session 2024-25 onwards)

B. Tech 3rd Semester: Instrumentation Engineering

Semester III

Sl. No.	Course Code	Course	Hours per Week			Credit	Marks	
			L	T	P	C	CE	ESE
Theory								
1	EE241301	Analog Electronics	3	0	0	3	30	70
2	IE241302	Fundamentals of Instrumentation & Transducers	3	0	0	3	30	70
3	EE241303	Digital Electronics	3	0	0	3	30	70
4	EE241304	Electric Circuit Analysis	3	1	0	4	30	70
5	MA241305C	Mathematics III-C	3	0	0	3	30	70
6	HS241306	Indian Knowledge Systems	2	0	0	2	-	50
7	AC241307	Language other than MIL	1	1	0	0 (PP/NP)	-	50
Practical								
1	EE241311	Analog Electronics Lab	0	0	2	1	15	35
2	IE241312	Transducers Lab	0	0	2	1	15	35
3	EE241313	Digital Electronics Lab	0	0	2	1	15	35
4	SI241321	Social Internship	0	0	4	2	-	50
TOTAL			18	2	10	23	195	605
Total Contact Hours per week: 30								
Total Credits: 23								

N.B. AC241307 is a Mandatory Audit Course (No Credit). It will be evaluated as PP(Pass) or NP (Not Pass)

DETAILED SYLLABUS

Course Code	Course Title	Hours per week L-T-P	Credit C
EE241301	Analog Electronics	3-0-0	3

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1** Explain the principles of various types of diodes and their applications.
- **CO2** Analyze the characteristics of BJTs and the performance of BJT amplifiers.
- **CO3** Analyze the performance of FETs and its various applications.
- **CO4** Examine the effects of negative and positive feedback on amplifiers.
- **CO5** Design OPAMP based applications.

MODULE 1: Junction Diodes

(6 Hours)

Review of PN junction diodes; rectifier circuits; clippers and clampers; Zener diode and voltage regulation, LED as photo-diode.

MODULE 2: Bipolar Junction Transistor (BJT)

(12 Hours)

Principle of operation of BJT: Transistor Currents, transistor characteristics- α and β parameters, early effect, Ebers-Moll model, amplification and switching action, transistor biasing circuits. Stability of operation and stability factor, small signal re-parameter and h-parameter models. Introduction to π -model, analysis of amplifiers with small signal models, current gain, voltage gain, power gain, input impedance and output impedance, cascaded amplifier, coupling between stages, frequency response and band width of an amplifier, Darlington connection, Differential amplifiers

MODULE 3: Field Effect Transistors

(8 Hours)

BJT vs FET

JFET: Basic construction and physical behaviour, current flow, Drain and Transfer Characteristics. Biasing circuits.

MOSFET: Operation in enhancement and depletion modes, small signal model of FET, analysis of FET as amplifiers (CS, CD, CG) and switches, introduction to CMOS.

MODULE 4: Feedback and Oscillators

(4 Hours)

Effect of negative feedback in amplifiers and its advantages; analysis of different feedback configurations; Barkhausen criteria for oscillation; analysis of RC, LC and crystal oscillators.

MODULE 5: Integrated Circuit Electronics

(10 Hours)

Internal circuit diagram of OP-AMPS, ideal characteristics of OPAMP, open-loop gain, input and output impedance, bandwidth, CMRR, slew rate, input offset voltage and current; non-inverting and inverting amplifiers, concept of virtual ground, adders, subtractors, comparators, integrators and differentiators, differential amplifiers and instrumentation amplifiers, OPAMP sinusoidal oscillators. Active Filters, LPF, HPF, BPF and notch filters, Introduction to Timer IC555 and its applications.

Textbooks/References:

1. Jacob Millman and Christos C Halkias, Integrated Electronics, PHI, Paperback 2017.
2. Ramakant A. Gayakwad, OP-AMPS and Linear Integrated Circuits, PHI, Fourth Edition (2021).
3. David J Bell (Oxford), Electronic Devices and Circuits, Oxford Fifth Edition (2008).
4. Adel S Sedra and Kenneth C Smith, Microelectronics Circuits, Oxford 2010.
5. NPTEL course as suggested by the department

Course Code	Course Title	Hours per week L-T-P	Credit C
IE241302	Fundamentals of Instrumentation & Transducers	3-0-0	3

Course Outcomes:

At the end of the course, the students will be able to

- **CO1** Analyze static and dynamic characteristics of a measurement system.
- **CO2** Understand the working principles of resistive, inductive, capacitive and piezoelectric transducers.
- **CO3** Demonstrate the working principles of transducers used for temperature measurement.
- **CO4** Select special transducers for measurement of various physical parameters.
- **CO5** Apply knowledge of Transducers and sensors for practical implementations in engineering applications.

MODULE 1: Introduction

(7 Hours)

Concepts and terminology of measurement system, transducer, sensor, classification of transducers, static and dynamic characteristics of measurement system, selection criteria, standards and calibration.

MODULE 2: Resistive transducers

(5 Hours)

Potentiometers: Principle, signal conditioning. Strain gauges: types, principle, signal conditioning circuits, strain gauge applications in measurement of pressure, force.

MODULE 3: Inductive transducers

(5 Hours)

Principle, signal conditioning, applications, etc. of LVDT & Synchros.

MODULE 4: Capacitive transducers

(5 Hours)

Types, signal conditioning circuits, applications in measurement of displacement and liquid level.

MODULE 5: Piezoelectric transducers

(5 Hours)

Piezoelectric crystal and its properties; Sensitivity coefficients; Materials, Applications in measurement of force, pressure, acceleration, etc.

MODULE 6: Temperature Measurement

(7 Hours)

Thermocouples: Thermoelectric effects, laws of thermocouple, cold junction compensation techniques, thermocouple types, construction, measuring circuits, signal conditioning circuits, RTD, Thermistors, IC temperature sensor.

MODULE 7: Special Transducers and introduction to advances in sensors technology

(6 Hours)

LDR, Radiation pyrometer, Hall effect transducers and Magneto strictive transducers.

Introduction to Smart sensors, MEMS, Nano sensors, Semiconductor sensors, Optical fiber sensors.

Text Books:

1. Ernest Doebelin, Measurement Systems: Application and Design, McGraw Hill. 5th Edition 2004.
2. John P. Bentley, Principles of Measurement Systems, Pearson Education. 4th Edition, 2005.
3. Hermann K. P. Neubert Instrument Transducers – An Introduction to their Performance and Design, Oxford, 1st Edition, 1963.
4. D.V. S Murthy, Transducers and Instrumentation, PHI, 2nd Edition, 2008.
5. D Patranabis, Sensors and Transducers, PHI, 2004

Course Code	Course Title	Hours per week L-T-P	Credit C
EE241303	Digital Electronics	3-0-0	3

Course Outcomes: At the end of this course, the students will be able to

- **CO1** Apply Boolean algebra to simplify logic expressions representing digital circuits.
- **CO2** Design a digital circuit using the minimum number of gates for a given logical expression.
- **CO3** Analyze and design of basic combinational circuits.
- **CO4** Analyze and design basic sequential circuits.
- **CO5** Use logic gates, programmable memory devices and A/D and D/A converters for practical circuit design.

MODULE 1: Digital fundamentals (8 Hours)

Representation of binary numbers, octal and hexadecimal numbers, complements, signed binary numbers, floating point numbers, binary codes, code conversion; Boolean expressions and logic gates, truth tables; laws of Boolean algebra, canonical and standard forms-SOP and POS; logic levels and pulse waveforms.

MODULE 2: Simplification of Boolean functions (6 Hours)

Karnaugh map, minimal SOP & POS, don't care conditions, multiple output minimization, Quine-McClusky method, determination and selection of prime implicants.

MODULE 3: Combinational Logic circuits (9 Hours)

Logic synthesis: universal property of NAND and NOR gates; Adders and subtractors, parallel binary adders, magnitude comparator, decoders and encoders, Multiplexer and demultiplexers, parity generators and checkers.

MODULE 4: Sequential Logic Circuits (9 Hours)

Flip Flops (FFs) and Latches, S-R, J-K, D and T FFs, Excitation table, Triggering of FFs & Latches, one shot astable multivibrator. Registers and counters: Definition, Modes of operation of registers and counters, Shift-Registers, Ripple Counters, Synchronous Counters, Ring Counters, Timing Sequences, Design Procedure.

MODULE 5: IC Structures, Digital memory, A/D and D/A converters (8 Hours)

Introduction to DTL, TTL, ECL, CMOS logic families; ROM, RAM and PLA; Fan-in, fan-out, power dissipation, propagation delay, figure of merit, noise level; A/D and D/A conversions: D/A Converter- Specifications- weighted resistor type, R-2R Ladder type. A/D Converters – Specifications – Flash type – Successive Approximation type.

Fan-Out, Power dissipation, Propagation delay, figure of merit, noise level

Textbooks/References:

1. M. Marris Mano, Digital Design, Pearson Edition of 2018.
2. N. N. Biswas, Logic Design Theory Prentice – Hall Edition of 1992.
3. T. L. Floyd, Digital Fundamental Pearson Edition of 2017.
4. R. P. Jain, Digital Electronics, Mc Graw Hill Edition of 2009.
5. NPTEL course as suggested by the department.

Course Code	Course Title	Hours per week L-T-P	Credit C
EE241304	Electric Circuit Analysis	3-1-0	4

Course Outcomes: At the end of the course, the students will be able to:

- CO1 Apply network theorems for solving electric circuits.
- CO2 Apply elementary graph theory for network analysis
- CO3 Analyze transient response of various electrical networks.
- CO4 Analyze Two-port network characteristics and parameters.
- CO5 Synthesize various networks.

MODULE 1: Network theorems for DC, AC and Reduction Circuits

Analysis of electrical networks with independent, dependent, ideal, practical voltage and current sources, Millman's theorem and Tellegen's theorem (AC and DC), reciprocity theorem, compensation theorem, Maximum Power Transfer Theorem, Thevenin's Theorem and Norton's Theorem for AC circuits; analysis of coupled circuits; dot rule and equivalent conductively coupled forms of magnetically coupled circuits.

MODULE 2: Network topology

Elementary Graph theory as applied to electrical networks. Matrices of graph: incidence matrix, tie-set matrix, circuit matrix, cut set matrix; advanced techniques of equation formulation for numerical solutions.

MODULE 3: Transient phenomenon

Initial and final conditions in network elements, forcing functions, step and ramp functions. Solution of simple circuits undergoing transient disturbances, A.C. transients, time domain equations and solutions by Laplace transforms.

MODULE 4: Two port Network

General principles, relationship of two port variables, impedance parameters, admittance parameters, transmission parameters, hybrid parameters. Relationship among parameters. π network, T network, interconnections of two-port networks.

MODULE 5: Network Synthesis

Realizability of networks, positive real function (PRF) and its properties, Hurwitz polynomial, Routh-Hurwitz Array, properties of various immittance functions (LC, RC, RL), Foster and Cauer forms of realization of network for a given driving point impedance/admittance function.

Text Books:

1. Abhijit Chakravarti, Circuit Theory Analysis and Synthesis, Dhanpat Rai & Comp 2021
2. A. Anand Kumar, Network Analysis and Synthesis, PHL Learning Pvt. Ltd 2019
3. J.A. Edminster, Electric Circuit McGraw Hill 2017
4. D Roy Chowdhury, Networks and Systems, New Age 2013
5. M.L Soni, A course in Electric Circuit Analysis, Dhanpat Rai Publication 2016
6. M E Van Valkenburg, Network Analysis, Pearson Education 2019
7. W. H. Hayt Jr. & J.E. Kemmerly, Engineering Circuit analysis, McGrawHill 2013
8. NPTEL course as suggested by the department

Course Code	Course Title	Hours per week L-T-P	Credit C
MA241305C	Mathematics III-C	3-0-0	3

Course Outcomes:

At the end of this course, the students will be able to

- **CO1** Apply Laplace transform techniques for solution of engineering problems
- **CO2** Solve the problems of vector spaces and Linear transformations
- **CO3** Solve engineering problems using numerical methods
- **CO4** Apply probability and statistical techniques for solution of engineering problems

MODULE 1: Laplace Transforms

(7 Hours)

Definition and properties of Laplace Transform and Inverse Laplace Transform, Laplace Transform and Inverse Laplace Transform of derivatives and integrals, Method of finding inverse Laplace Transform using partial fractions, Convolution Theorem, Solution of ordinary differential equations using Laplace Transforms.

MODULE 2: Linear Algebra

(7 Hours)

Linear Systems of Equations: Existence and Uniqueness of solutions and Dimension of the Solution Space associated with the system of linear equations; Vector Spaces: Definition of Vector Space, Sub-spaces, Linearly Independence and Dependence, Linear Span, Basis, Dimension

Row, Line and Column space of a matrix

Basis and Dimension: Definition and examples, Linear dependence and independence, Dimension of a subspace. Linear transformations: Definition and examples, the Kernel of a Linear Transformation, Matrix representation of linear transformation and systems of linear equations, Rank-nullity theorem.

MODULE 3: Numerical Methods

(14 Hours)

Errors in Numerical Calculations, Solution of Algebraic and Transcendental Equations: Bisection Method, Iteration Method, Newton-Raphson Method. Interpolation: Newton Forward, Backward and Lagrange interpolation, Numerical Differentiation and Integration: Trapezoidal and Simpson's rule. Solution of Linear Systems: Gauss elimination and pivoting, Gauss-Seidel methods, Numerical Solution of Ordinary Differential Equations: Euler's and Runge-Kutta Methods.

MODULE 4: Probability and Statistics

Basic Probability Theorems, Conditional Probability and Baye's theorem; Random Variables, Probability Distributions; Mean and Variance, Median and Mode; Binomial, Poisson, Gaussian and Normal distribution and their applications; Joint distribution: Covariance and Correlation; Random Sampling, Estimation of Parameters: Point Estimate, Interval Estimate, Maximum Likelihood Method; Confidence Intervals, Testing of Hypothesis, Goodness of Fit Test, Curve Fitting: Least Square Method.

Text Books:

1. Kreyszig, E., Advanced Engineering Mathematics, John Wiley, 2011.
2. Strang G., Wellesley MA, Introduction to Linear Algebra, Cambridge Press, 2016.
3. Jain M.K., Iyengar, S. R. K., Jain, R. K., Numerical Methods for Scientific and Engineering Computation New Age International (P) Ltd, 8th Edition, 2022.
4. Ross, S., Introduction to Probability and Statistics for Engineers and Scientists, Pearson 6th Edition, 2021.

Course Code	Course Title	Hours per week L-T-P	Credit C
HS241306	Indian Knowledge Systems	2-0-0	2

Course Objectives:

1. To introduce students to the foundational concepts of Indian Knowledge Systems (IKS).
2. To explore the contributions of ancient and medieval India in science, technology, mathematics, and philosophy.
3. To develop an appreciation for traditional Indian knowledge and its relevance in modern contexts.
4. To encourage critical thinking about indigenous knowledge systems and their applications.

Course Outcome (CO):

After completion of the course, the students will be able to

- **CO1:** Understand the key principles of Indian Knowledge Systems.
- **CO2:** Recognize India's contributions to science, mathematics, medicine, and engineering.
- **CO3:** Analyze the interdisciplinary nature of IKS and its connections with modern technology.
- **CO4:** Appreciate the philosophical and ethical dimensions of traditional Indian knowledge.

MODULE 1: Introduction to Indian Knowledge System (IKS) (4 Lectures)

- Definition, scope, and significance of IKS
- Sources of Indian knowledge: Vedas, Upanishads, Puranas, Smritis, and other texts
- Classification of knowledge in Indian tradition (Vidyas and Kalas)

MODULE 2: Science & Technology in Ancient India (12 Lectures)

- Contributions to mathematics (Zero, Decimal System, Vedic Mathematics, Aryabhata, Brahmagupta)
- Astronomy & Space Science (Surya Siddhanta, Planetary Motion, Concept of Time)
- Metallurgy and material sciences (Iron Pillar, Wootz Steel, Ayurvedic alloys)
- Civil engineering & architecture (Temple architecture, Town planning, Water management)

MODULE 3: Indian Philosophical & Ethical Systems (6 Lectures)

- Key schools of Indian philosophy (Nyaya, Vaisheshika, Samkhya, Yoga, Vedanta)
- Concept of Dharma, Karma, and sustainable living
- Ethics in science and technology from an Indian perspective

MODULE 4: Traditional Medicine & Wellness Systems (4 Lectures)

- Foundations of Ayurveda, Yoga, and Siddha
- Holistic health and wellness practices
- Relevance of traditional medicine in modern healthcare

MODULE 5: IKS in Contemporary Applications (4 Lectures)

- Revival of traditional knowledge in modern science and engineering
- Case studies (e.g., Panchakarma in wellness tech, Vastu in architecture)
- Government initiatives (National Education Policy 2020, IKS Division under AICTE)

Text books:

1. "Indian Knowledge Systems" – Kapil Kapoor & Avadhesh Kumar Singh
2. "Ancient Indian Leaps into Mathematics" – B.S. Yadav
3. "Science and Technology in Ancient India" – O.P. Jaggi
4. "The Positive Sciences of the Ancient Hindus" – Brajendranath Seal
5. AICTE Model Curriculum for IKS (Available online)

Online Resources:

- NPTEL / SWAYAM courses on IKS
- AICTE-IKS Portal (<https://iksindia.org/>)
- Digital Library of India (<https://dli.gov.in>)

Course Code	Course Title	Hours per week L-T-P	Credit C
AC241307	Language other than MIL	1-1-0	0

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Demonstrate a clear understanding of basic grammar rules including subject-verb agreement, tenses, and cases, and apply them in constructing grammatically correct sentences.
- **CO2:** Identify and correct common grammatical and usage errors in written English, enhancing sentence effectiveness and clarity.
- **CO3:** Enhance their vocabulary through the use of synonyms, antonyms, idiomatic expressions, and phrasal verbs, improving both written and spoken communication.
- **CO4:** Apply verbal reasoning skills to comprehend, arrange, and complete sentences and paragraphs effectively.
- **CO5:** Develop strong reading comprehension skills using skimming, scanning, and intensive reading techniques to extract key information from various texts.
- **CO6:** Produce coherent and concise writing through precis writing and essay composition, demonstrating logical flow, clarity, and appropriate structure.

MODULE 1: Basic grammar usage

(10 Lectures)

Subject-verb agreement, narration, tenses

Case: Subjective case, possessive case, objective case

Writing effective sentences, Spotting errors

MODULE 2: Vocabulary Building

(6 Lectures)

Synonyms and Antonyms, One word substitution,

Idioms and Phrases and Phrasal Verbs

MODULE 3: Verbal ability

(8 Lectures)

Cloze test

Sentence rearrangement, Sentence completion,

Paragraph Jumbles

MODULE 4: Writing and Reading Ability

(8 Lectures)

Reading subskills, Skimming, Scanning and Intensive reading

Reading Comprehension, Precis Writing, Essay Writing

Text Books

1. Objective General English: S.P.Bakshi, Arihant Publication India Ltd., 2021
2. Descriptive English: S.P.Bakshi, Arihant Publication India Ltd., 2016

Course Code	Course Title	Hours per week L-T-P	Credit C
EE241311	Analog Electronics Lab	0-0-2	1

Course Outcomes: After completing this course, the students will be able to:

- **CO1** Acquire practical knowledge of solid-state electronics including diodes, BJT, FET and operational amplifiers.
- **CO2** Develop the ability to analyze and design analog electronic circuits using discrete components.
- **CO3** Design, construct and take measurements of various analog circuits to compare experimental results in the laboratory with theoretical analysis.

LIST OF EXPERIMENTS

Part I (Virtual Lab)

1. Zener diode characteristics and Zener as a voltage regulator
2. Input and output characteristics of transistor CB configuration
3. Input and output characteristics of transistor CE configuration
4. Studies on BJT CE amplifier
5. Studies of JFET, MOSFET characteristics
6. Studies of basic properties of Operational Amplifier: Inverting and Non-Inverting Amplifiers

Part II (Hardware Experiments)

1. Implement rectifiers using pn junction diodes and study the effect of filters
2. Input and output characteristics of a BJT CB configuration
3. Input and output characteristics of a BJT CE configuration
4. Studies on BJT CE amplifier
5. Studies of differentiator and integrator using Operational Amplifier

Course Code	Course Title	Hours per week L-T-P	Credit C
IE241312	Transducers Lab	0-0-2	1

Prerequisites: Fundamentals of Instrumentation and Transducer

Course Outcome:

After completing this course, the students will:

- **CO1** Study the principles of different types of transducers.
- **CO2** Study the applications of different types of transducers.
- **CO3** Interpret the experimental results and draw meaningful conclusions.
- **CO4** Develop laboratory skill in handling various measuring instruments.

LIST OF EXPERIMENTS

1. To study the operating principle of RTD and measure temperature by RTD.
2. To study the operating principle of Thermistor and measure temperature by Thermistor.
3. To study the operating principle of Thermocouple and measure temperature by Thermocouple.
4. To study the operating principle of Strain Gauge and measure Strain by Strain Gauge.
5. To study the operating principle of LVDT and measure displacement by LVDT.
6. To study the operating principle of Capacitive Transducer and measurement of linear and angular displacement by Capacitive Transducer.
7. To study the operating principle of LDR and measure light intensity by LDR.
8. To study the operating principle of Inductive Transducer and measure displacement by Inductive Transducer.

Suggested Textbooks/References:

1. Ernest Doebelin, Measurement Systems: Application and Design, McGraw Hill. 5th Edition 2004.
2. John P. Bentley, Principles of Measurement Systems, Pearson Education, 4th Edition, 2005.
3. Hermann K.P. Neubert, Instrument transducers – An introduction to their performance and design, Oxford 1st Edition 1963.
4. D.V.S Murthy, Transducers and Instrumentation, PHI, 2nd Edition 2008.
5. D Patranabis, Sensors and Transducers, PHI, 2004.

Course Code	Course Title	Hours per week L-T-P	Credit C
EE241313	Digital Electronics Lab	0-0-2	1

Course Outcomes:

At the end of the course, the students will be able to:

- **CO1** Realize basic gates and verify their truth tables
- **CO2** Perform experimental analysis of combinational and sequential logic circuits.
- **CO3** Write an engineering report in an organized way

LIST OF EXPERIMENTS

1. Realization of basic gates by using universal gates and verification of their truth tables.
2. Realization of half adder and full adder circuits by using basic gates.
3. Combinational Logic Design using 74xx ICs
4. Arithmetic Circuit construction and testing using 74xx ICs: Adders
5. Construction of 1-bit comparator using 74xx ICs.
6. Code converters –Binary to Gray and Gray to binary using basic gates.
7. Verification of Truth Table of SR Flip-Flop using NOR and NAND gates.
8. Verification of Truth Tables of JK, D, T Flip-Flops using 74xx ICs
9. Construction of 3-Bit synchronous counter using flip flops.
10. Construction of Decade Counter using 74xx ICs.

Suggested Books and Other Study Materials

1. M. Marris Mano, Digital Design, Pearson, 1984.
2. R. P. Jain, Digital Electronics Practice using Integrated Circuit, McGraw Hill Education, 2009.
